V.E Backend Repository Management and Service-Level Trade-offs

[placeholder] The refined repository holds 942 entries against 6,000, and the cost of producing it is paid once while the cost of querying it is paid on every request. We keep the two ledgers separate and report neither an end-to-end latency nor an amortisation claim. *[placeholder] [placeholder]* The protocol indexes each repository on its own and times the same 55 questions, so the only quantity that changes between the two columns of Table 1 is how many experiences the retriever scans. *[placeholder]*

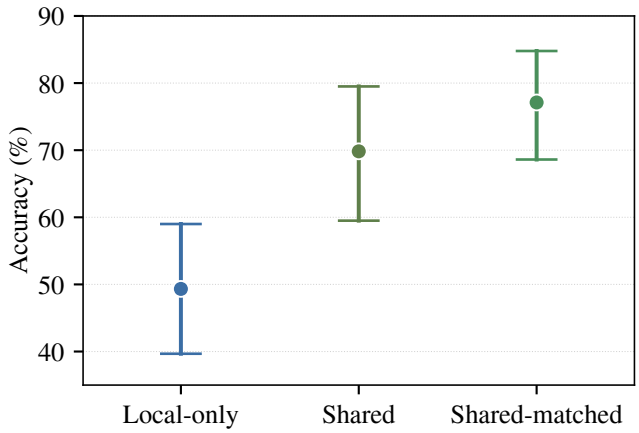


Figure 1: V.C figure kept on this page only to check the balance against the table. Accuracy on the 60 construction questions when the answering client holds no experience about them.

[placeholder] Producing the repository costs 94 s of similarity-matrix construction plus 0.38 s of anchor grouping, and the historical log records 275 uncached condensation requests. *[placeholder] [placeholder]*

Table 1: Backend management overhead on MMLU High School Geography: 6,000 entries in the full repository versus 942 after refinement ($\tau_{\text{sim}} = 40$, 84.3 % fewer). Local timings exclude file I/O and LLM inference; query statistics are the median of three measurement sessions (825 timed queries per repository). API rows summarise the historical uncached condensation requests recorded in the run log.

Metric	Full	Refined
Indexed entries	6,000	942
Index build (s)	0.0354	0.0079
Hot query P50 (ms)	22.67	3.60
Hot query P95 (ms)	37.72	5.73
BM25 matrix build (s)	–	94.1
Anchor grouping (s)	–	0.38
Uncached LLM calls	–	275
Input / output tokens	–	186,279 / 12,864
Request P50 / P95 (s)	–	1.691 / 4.052
Logged cost (USD)	–	0.0103

[placeholder] Query-side savings are measured, not inferred: the index shrinks from 6,000 to 942 entries, construction drops from 0.035 s to 0.008 s, and the median query falls from 22.67 ms to 3.60 ms while the 95th percentile falls from 37.72 ms to 5.73 ms. *[placeholder] [placeholder] [placeholder]*

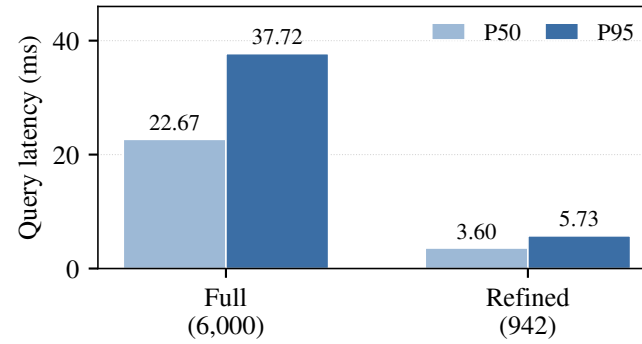


Figure 2: Optional companion: full vs refined hot-index query latency, P50 and P95 on one axis, median of three measurement sessions.

[placeholder] The optional figure repeats the table at a coarser grain and can be dropped if the page is tight. *[placeholder] [placeholder]*